

An Overview of the Proposed PE Study

Main architecture, staged training, and ROI ablation plan

– – Quoc Duy – –

1. Study scope: two PE tasks

- Dataset (evaluation using 3-fold cross-validation):
 - + STANFORD INSPECT data (**train/validation/internal test**)
 - + Turkey (**external test only**)
 - (Radiology reports in **STANFORD INSPECT** used only during training for image-report pretraining and silver-label generation)
- Two Complementary Tasks for Pulmonary Embolism:

Task A: PE Diagnosis

Objective: Detect pulmonary embolism and characterize its imaging phenotype.

Inputs:

- + Full 3D CTPA volume
- + Anatomy-aware features from the heart, pulmonary arteries, and lungs

Expected outputs:

- + **PE presence: positive / negative (priority) - main task**
- + PE acuity: acute / chronic / Acute-on-chronic/ uncertain
- +PE location attributes: central, lobar, segmental, subsegmental and saddle involvement.
- +Right-heart strain findings: RV enlargement, RV/LV-related signs, septal bowing, or reflux
- + Optional associated thoracic findings

Task B: Multimodal PE Prognosis

Objective: Estimate the clinical risk and short-term outcome of patients with acute PE.

Inputs:

- + Full 3D CTPA and anatomy-aware image features
- + Clinical and EHR variables
- + Missingness indicators
- + PESI or sPESI score

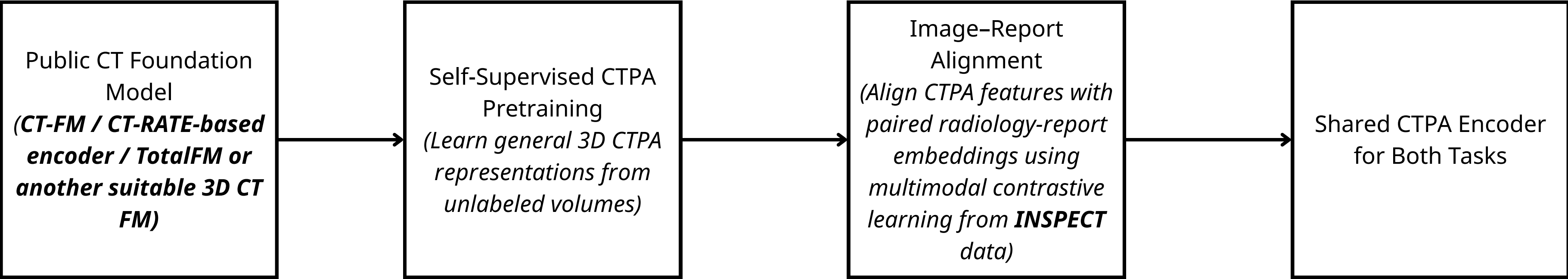
Expected outputs:

- Primary: 30-day mortality in confirmed acute PE patients.
- Secondary: 30-day mortality in the all-comer CTPA cohort.
- Exploratory: 6-/12-month mortality, readmission and 12-month PH.

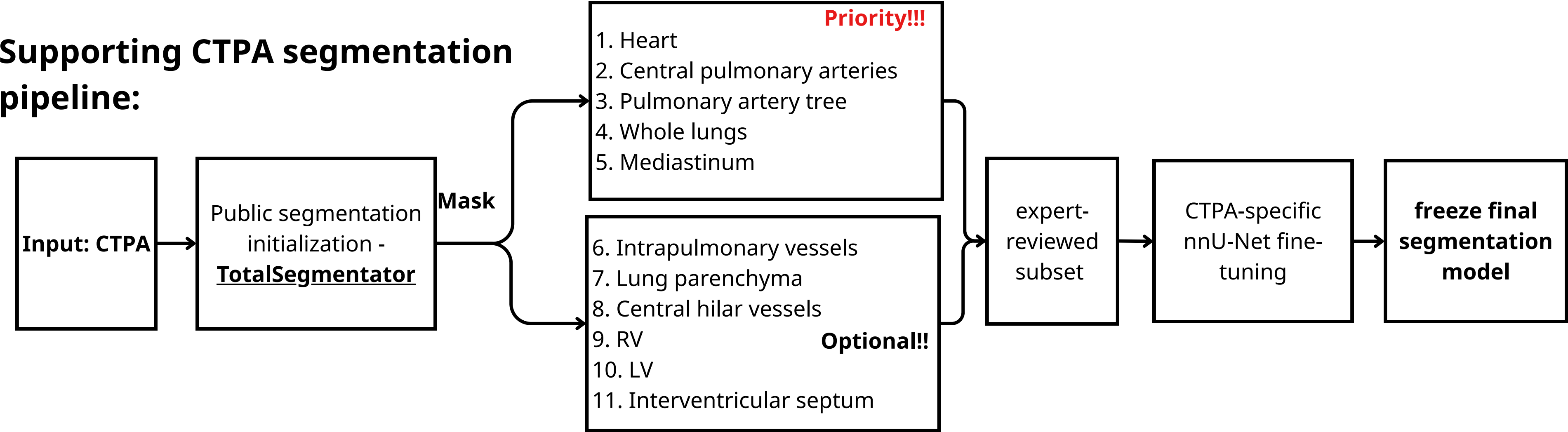
Native labels are used where available. Report-derived silver labels are used as auxiliary training targets and are evaluated only when gold or expert-reviewed annotations are available.

2. Shared foundation before task-specific models

Model PE Task:

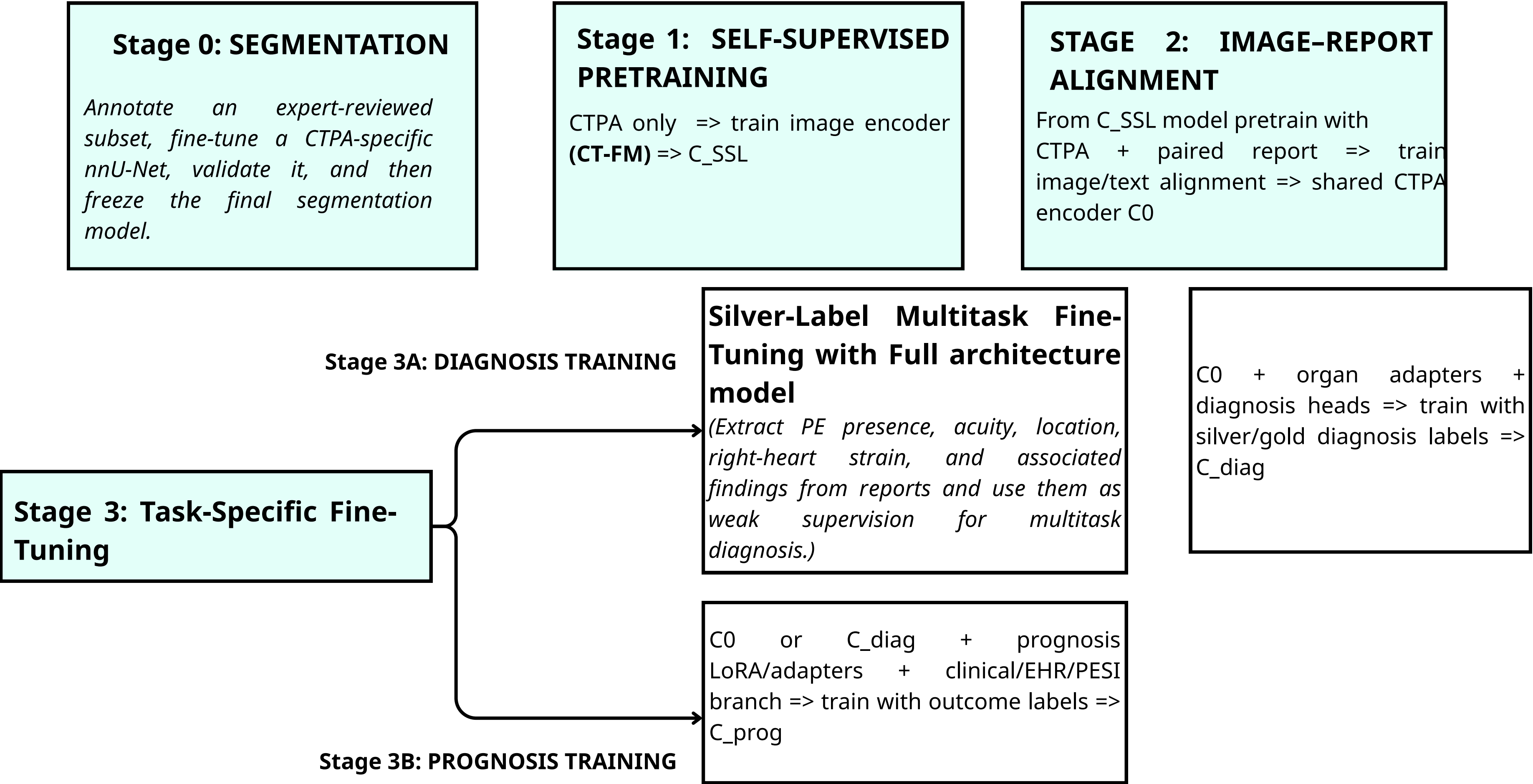


Supporting CTPA segmentation pipeline:

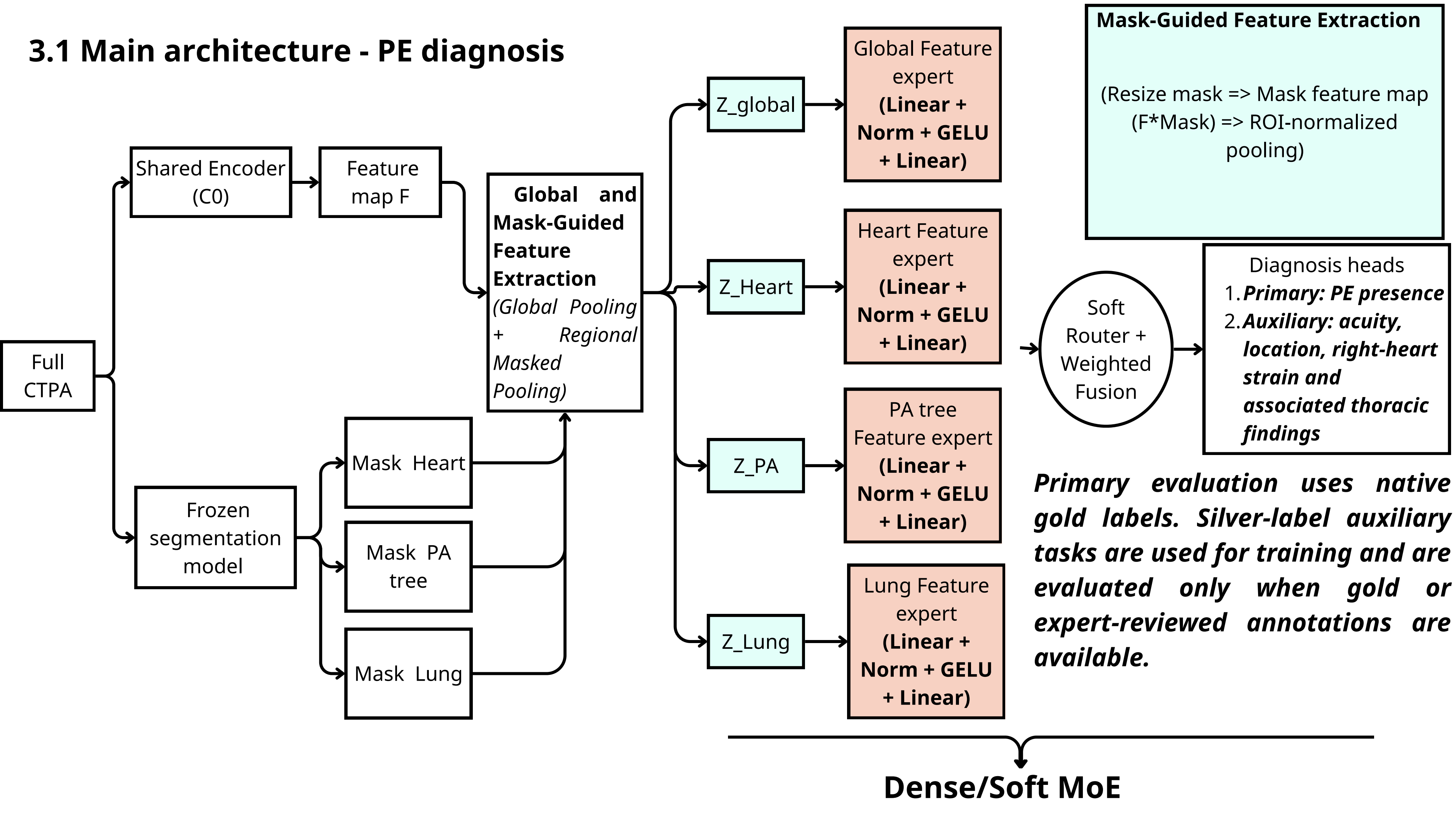


3. Main Architecture

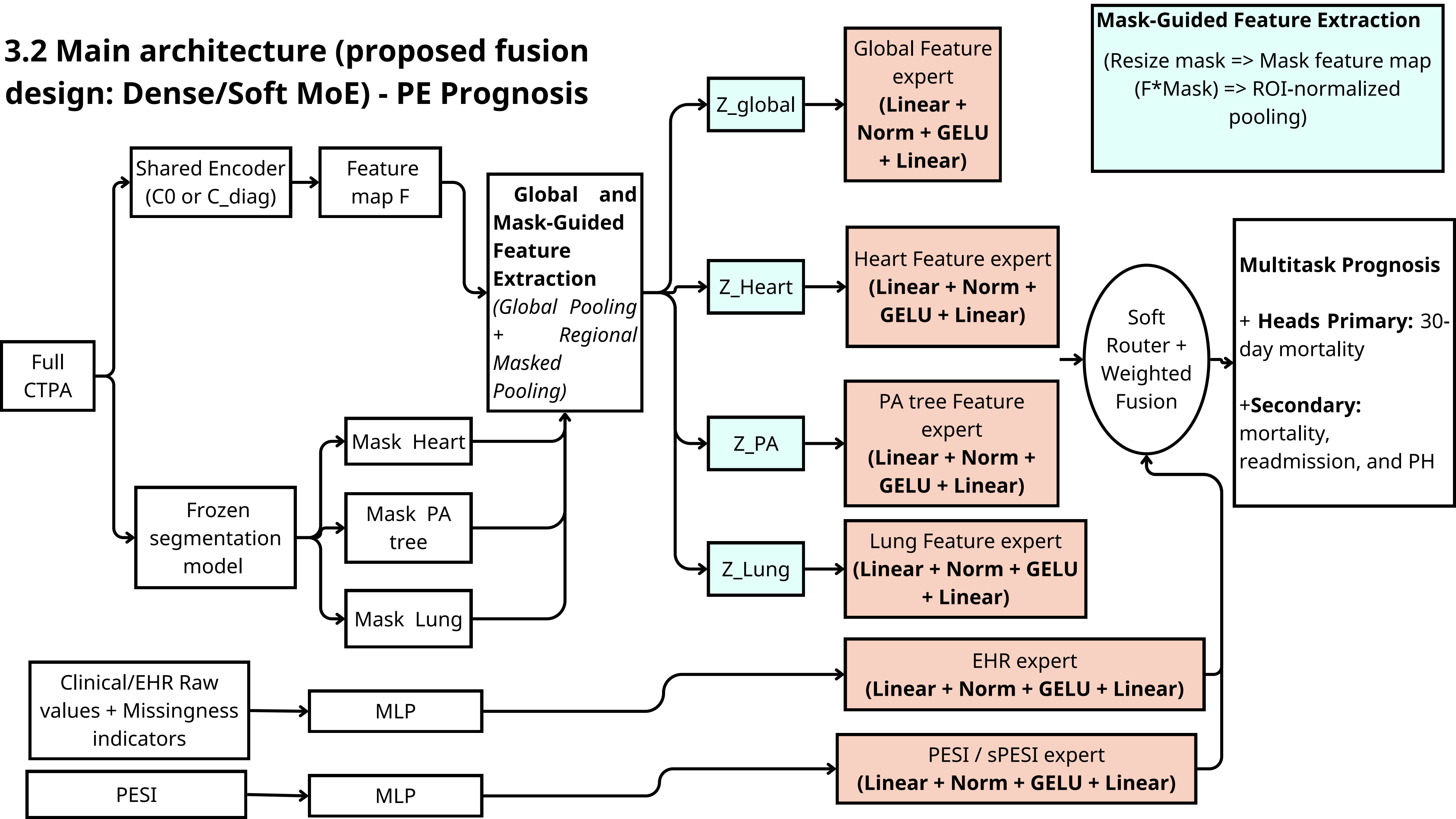
In task Prognosis using pretrain from encoder shared or weight of task diagnosis



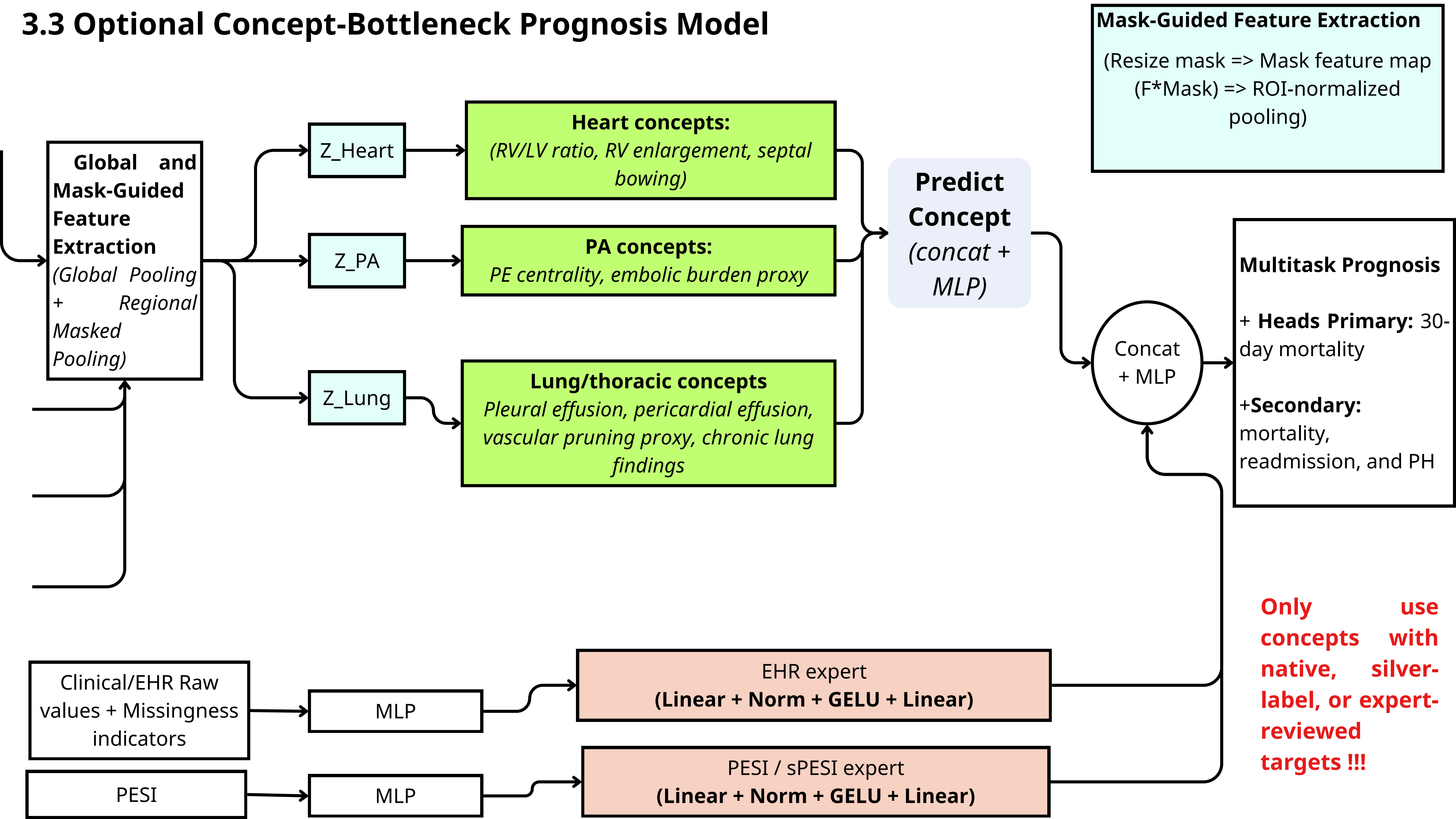
3.1 Main architecture - PE diagnosis



3.2 Main architecture (proposed fusion design: Dense/Soft MoE) - PE Prognosis



3.3 Optional Concept-Bottleneck Prognosis Model



3.4 Evaluation Protocol

Diagnosis

- Patient-level stratified three-fold CV.
- AUROC, AUPRC, sensitivity, specificity, F1.
- Internal testing on INSPECT.
- External testing on Turkey.
- Bootstrap 95% confidence intervals.
- STARD+AI reporting.

Prognosis

- Primary cohort: confirmed acute PE.
- Repeated or nested patient-level CV depending on event count.
- AUROC, AUPRC, sensitivity and specificity.
- Calibration curve, calibration slope/intercept and Brier score.
- Bootstrap 95% confidence intervals.
- Paired comparison on the same patients.
- TRIPOD+AI reporting.
- External testing on Turkey (if data having labels and clinical variables).

A patient-level temporal hold-out will be preferred when reliable acquisition dates and sufficient outcome events are available; otherwise, patient-level stratified three-fold cross-validation will be used.

4. Ablation study: three experiment families (in two tasks)

Ablation 1 Architecture ablation

Retrain variants from C0: remove expert branches, remove router, remove multitask labels. Tests component contribution.

Ablation 2 Remove-ROI counterfactual

Do not retrain. Test C_full on full CT versus remove-heart / remove-PA / remove lung / random. Tests necessity.

Ablation 3 ROI-only students

Train student models from C0 on heart-only, PA-only, lung-only, random. Tests sufficiency.

Checkpoint rule

- **Architecture ablation:** Train each variant from C0
- **Remove-ROI:** use C_full; no training
- **ROI-only student:** initialize from C0; optionally distill from C_full

ROI definitions: derive experimental ROIs from basic masks

Basic Masks
Heart, Mediastinum, PA tree, central PA, lung, lung vessels, hilar vessels

Keep-only ROI
Keep ROI intensity; replace everything outside ROI.
 $x_{\text{only}} = x \times M + c \times (1 - M)$

Remove ROI
Replace intensity inside ROI; keep the rest of CTPA unchanged
 $x_{\text{remove}} = x \times (1 - M) + c \times M$

Random control
Match random region inside body mask; same approximate volume

ROI	Meaning	Mask Operation	Experiment
ROI1	Heart + mediastinum	$M_{\text{heart}} \cup M_{\text{mediastinum}}$	Keep-only
ROI2	Strict heart	M_{heart}	Keep-only
ROI3	Exclude central PA	$M_{\text{centralPA}}$	Remove-roi
ROI4	Pulmonary arteries only	$M_{\text{PA_tree}}$	Keep-only
ROI5	Lung with vessels	M_{lung}	Keep-only
ROI6	Parenchyma only	$M_{\text{lung}} - M_{\text{vessels}}$	Keep-only
ROI7	Remove heart + hilum	$(M_{\text{heart}} \cup M_{\text{hilar}})$	Remove-roi
ROI8	Random crop	matched random mask	control

4.0 Baseline Comparisons

Diagnosis baselines

Model	Input	Main target
Report-only NLP baseline	Report	PE presence
PENet baseline	CTPA	PE presence
Full-CT image baseline	CTPA	PE presence
Full-CT image + FM initialization	CTPA	PE presence
Proposed anatomy-aware model	CTPA + masks	PE presence

Training variant	Labels
Single-task	Native pe_positive
Limited native multitask	pe_positive + pe_acute + pe_subsegmentalonly
Silver-label multitask	Native label + report-derived auxiliary labels

Prognosis baselines

Model	Input
PESI/sPESI-only baseline	PESI/sPESI
Clinical-only baseline	EHR + missingness indicators
Image-only Full-CT baseline	CTPA
Image + Clinical baseline	CTPA + EHR
Image + Clinical + PESI baseline	CTPA + EHR + PESI/sPESI
Proposed anatomy-aware multimodal model (ours)	CTPA + masks + EHR + PESI/sPESI

Multitask

Single-task PE presence
Native-label multitask
Silver-label multitask

4.1 Ablation 1: Architecture ablation

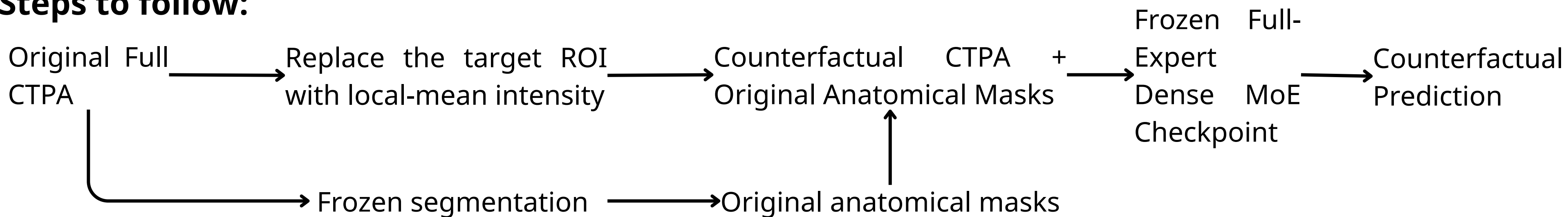
Global Only	Global + Heart	Global + PAtree	Global + Lung	Full Experts with Dense MoE	Full Experts w/o Router (Concatenation + Fusion MLP)	EHR only (<i>task prognosis</i>) 1. <u>all variables</u> 2. <u>Common-variable</u> 3. <u>Raw values w/o missingness indicators.</u> 4. <u>Raw values plus missingness indicators</u>	PESI/sPESI only (<i>task prognosis</i>)
-------------	----------------------	-----------------------	---------------------	---	---	---	---

4.2 Ablation 2: Remove-ROI counterfactual

- Use the frozen Full-Expert Dense MoE checkpoint. Replace the target ROI or a size-matched random region with local-mean intensity, preserve the original anatomical masks, and perform inference without retraining.
- Initial priority subset: strict heart, PA-only, lung-parenchyma-only and target-matched random controls. The complete ROI1 - ROI8 suite will be considered after confirming segmentation feasibility.

Full CTPA	Remove PA tree	Remove Heart	Remove Lung	Remove Size-Matched Random ROI 1.Remove Heart-matched Random ROI 2.Remove PA-matched Random ROI 3. Remove Lung-matched Random ROI
-----------	----------------	--------------	-------------	--

Steps to follow:

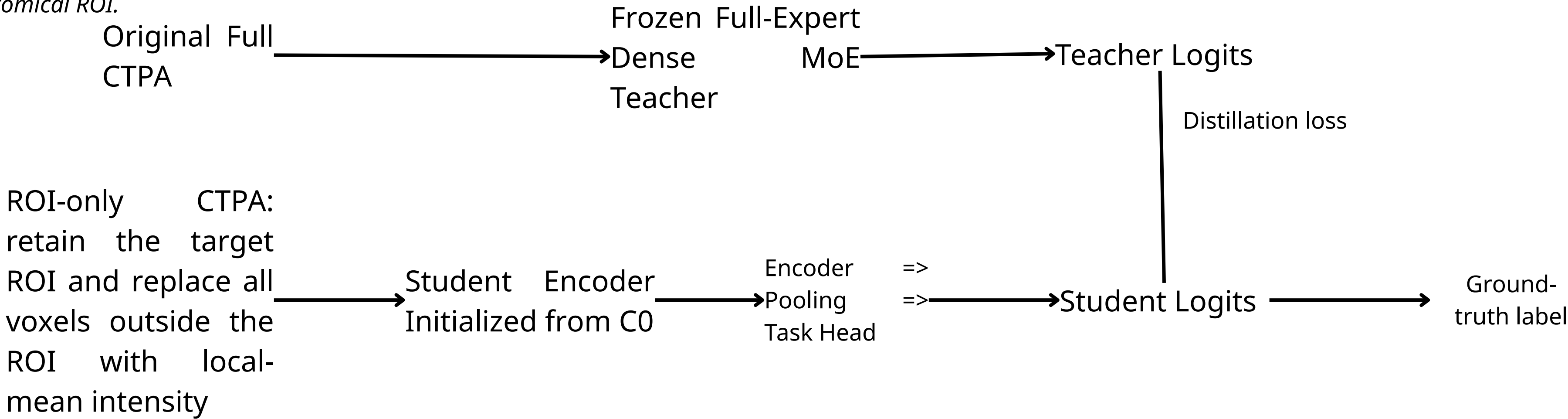


4.3 Ablation 3: ROI-only students

The trained Full-Expert Dense MoE model is frozen and used as the teacher. For each target anatomy, an ROI-only input is generated by retaining the selected ROI and replacing all voxels outside it with local-mean intensity. Each student is initialized from the same C0 checkpoint and trained separately using supervised task loss and optional knowledge distillation.

Heart student	PA tree student	Lung-parenchyma-only student	Matched Random ROI Student 1. Heart-size Matched Random-ROI Student 2. PA-size Matched Random-ROI Student 3. Lung-size Matched Random-ROI Student
----------------------	------------------------	-------------------------------------	---

Random control: A target-specific matched Random-ROI Student retains a random region with approximately the same volume as the corresponding anatomical ROI.



5. Summary

- Main architecture: one full CTPA + predicted masks => shared encoder => organ features => fusion.
- Diagnosis: image-centered multitask PE prediction; reports supply silver labels, not inference input.
- Prognosis: image experts + clinical/EHR/PESI; freeze most of the encoder and train PEFT/fusion/head.
- Architecture ablation: retrain variants from C0 to test model components.
- Remove-ROI: no retraining; test C_full after anatomically erasing regions.
- ROI-only students: train separate students from C0 to test whether each ROI is sufficient.

Scientific goal:

1. Determine which anatomical regions contain predictive information for diagnosis and prognosis, while distinguishing biologically plausible signals from acquisition or protocol-related confounding.
2. Strong performance from an ROI does not by itself establish a causal biological mechanism.

6. Question

Q1. Organ-specific adaptation

Is one shared full-CTPA encoder with mask-guided ROI pooling and post-encoder organ-specific feature experts sufficient, or should organ-specific LoRA/adapters be inserted within the shared encoder?

Q3. Prognosis architecture

Should Figure 3.2 be the primary prognosis model and Figure 3.3 an optional concept-based interpretability experiment, or should the concept-bottleneck architecture replace the direct multimodal model?

Q2. Mask source and validation

Should the primary downstream experiments use predicted masks from the frozen segmentation model, with ground-truth masks used only for oracle evaluation on an annotated subset? Is expert review plus quantitative segmentation validation required before freezing the model?

Q4. ROI and segmentation feasibility

Are the proposed ROIs clinically appropriate and technically feasible? Should one multi-class CTPA segmentation model generate all masks, or should multiple models and expert annotations be combined? Should we start with strict-heart, PA-only, lung-parenchyma-only and matched-random controls, or the complete ROI1–ROI8 suite?

7. Next Steps

1. Preprocess INSPECT data.
2. Develop and validate the CTPA segmentation model; save predicted masks for expert verification.
3. Evaluate public CT foundation-model initialization candidates: CT-FM, CT-RATE/CT-CLIP and TotalFM.
4. Develop and validate the report-mining pipeline for silver-label generation.
5. Continue in-domain CTPA self-supervised pretraining and image - report alignment.
6. Run diagnosis/prognosis baselines, proposed models, and counterfactual experiments.
7. Evaluate Turkey.

THANK YOU